

FRIDA

Frida LAB 8

Objective

In Challenge 0x8, the application validates the result of a method with parameters. Unlike previous challenges, you must hook a method that accepts one or more arguments and modify either the arguments or the return value to satisfy the application's check.

This challenge introduces hooking methods with parameters using Frida.

Install the APK

Install the application on your rooted Android device or emulator.

adb install FridaLab.apk

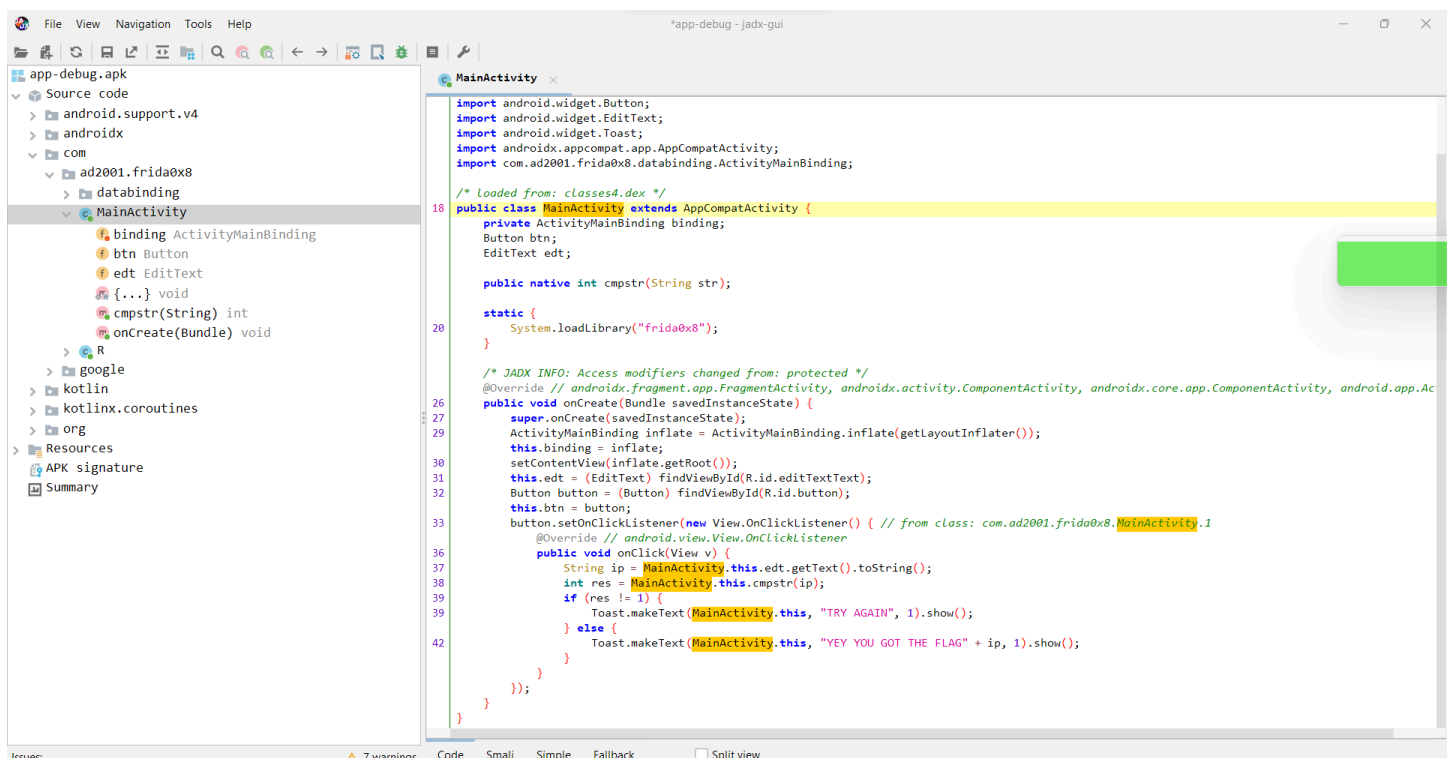
Launch the Application

Decompile the APK

Open the APK in JADX.

jadx-gui FridaLab.apk

Browse the source code.



At the beginning we see the declaration of the native function `cmpstr`.

```
public native int cmpstr(String str);
```

```
static {
```

```
    System.loadLibrary("frida0x8");
```

```
}
```

The `cmpstr` function takes a string as its argument and returns an integer.

It's loading the native library `frida0x8` into the memory. Let's see what's happening in the callback function of the button.

```
button.setOnClickListener(new View.OnClickListener() { // from class:  
com.ad2001.frida0x8.MainActivity.1
```

```
    @Override // android.view.View.OnClickListener
```

```
    public void onClick(View v) {
```

```
        String ip = MainActivity.this.edt.getText().toString();
```

```
        int res = MainActivity.this.cmpstr(ip);
```

```
        if (res == 1) {
```

```
            Toast.makeText(MainActivity.this, "YEY YOU GOT THE FLAG " + ip, 1).show();
```

```
        } else {
```

```
            Toast.makeText(MainActivity.this, "TRY AGAIN", 1).show();
```

```
        }
```

```
    }
```

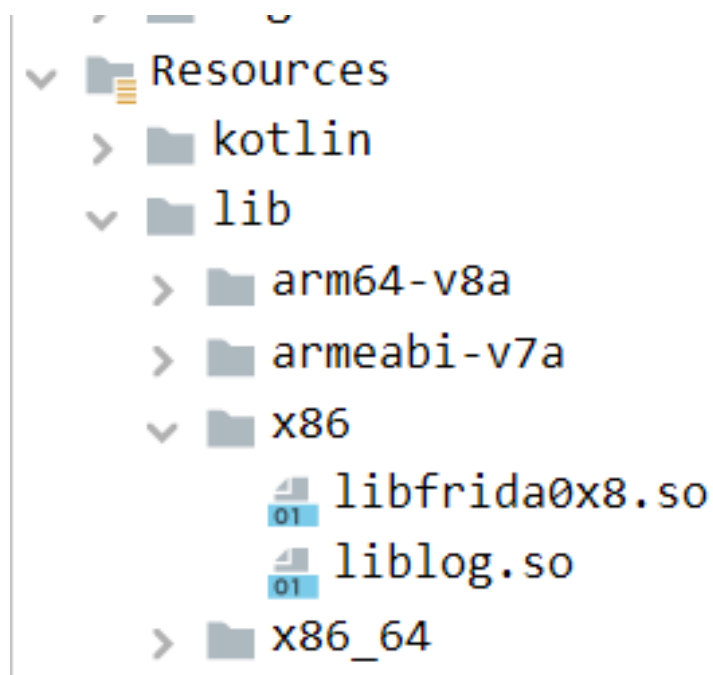
```
});
```

We can see that in the callback function, the application calls a method named `cmpstr` with the text from the `EditText` and returns an integer value. Another important thing is that our input is the flag itself. So let's conclude what we know so far.

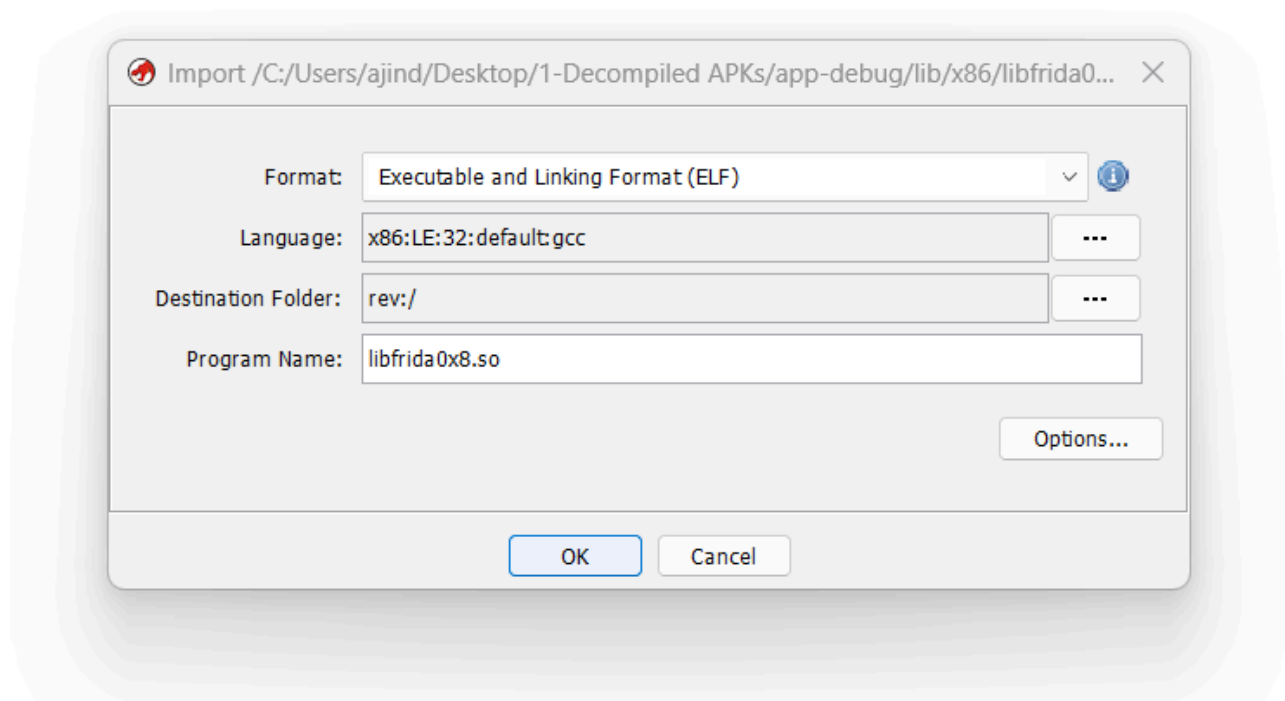
- The application dynamically loads a library named `frida0x8` into memory at runtime.
- The input is the flag itself.
- This input is then passed to a function called `cmpstr` within the `frida0x8` library.
- If the return value is `1`, we will get the flag.

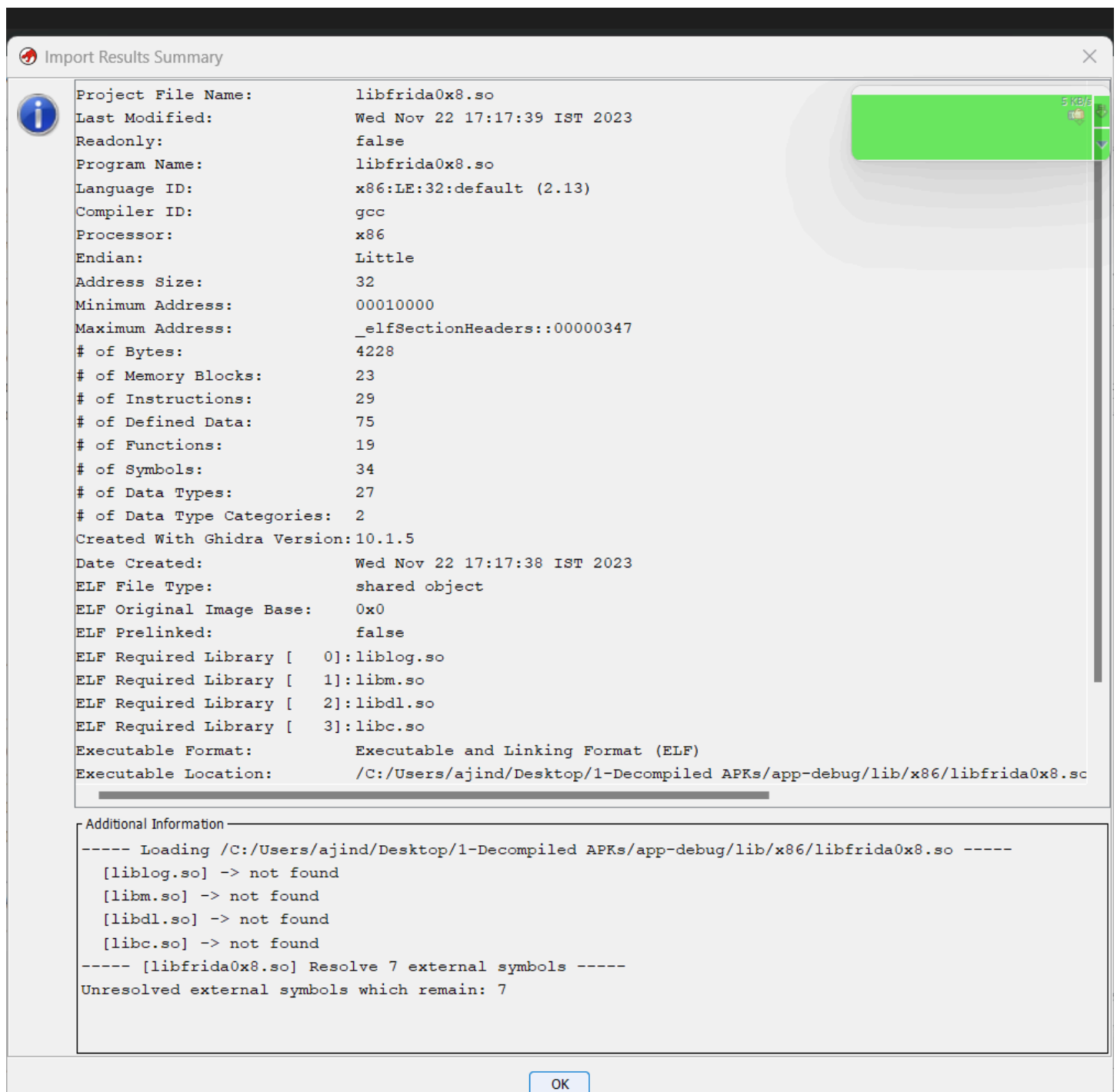
Let's now examine the frida0x8 library.

We can see where the library is at in jadx. It's under /resources/lib/



To reverse engineer and analyze this libfrida.so file we will be using a tool called ghidra.





The right panel displays the decompilation. If this is your first time reversing a native library, don't worry things get easier with practice. This time i will provide the source code for the native function for the sake of explanation.

```
#include <jni.h>
```

```
#include <string.h>
```

```
#include <stdio.h>
```

```
#include <android/log.h>
```

```
extern "C"
```

```
JNIEXPORT jint JNICALL
```

```
Java_com_ad2001_frida0x8_MainActivity_cmpstr(JNIEnv *env,  
object thiz, jstring str) {
```

```
    const char *inputStr = env->GetStringUTFChars(str, 0);
```

```
    const char *hardcoded = "GSJEB|OBUJWF`MBOE~";
```

```
    char password[100];
```

```
    for (int i = 0; i < strlen(hardcoded) ; i++) {
```

```
        password[i] = (char)(hardcoded[i] - 1);
```

```
    }
```

```
    password[strlen(hardcoded)] = '\0';
```

```
    int result = strcmp(inputStr, password);
```

```
    __android_log_print(ANDROID_LOG_DEBUG, "input ",  
"%s",inputStr);
```

```
    __android_log_print(ANDROID_LOG_DEBUG, "Password",  
"%s",password);
```

```
    env->ReleaseStringUTFChars(str, inputStr);
```

```
// Returning result: 1 if equal, 0 if not equal  
return (result == 0) ? 1 : 0;  
}
```

```
D:\Downlaods> adb shell
```

```
generic_x86:/ # cd /data/local
```

```
local.prop local/
```

```
generic_x86:/ # cd /data/local/tmp/
```

```
generic_x86:/data/local/tmp # ls
```

```
generic_x86:./frida
```

Run Frida

Start the application with Frida.

```
frida -U -f uk.rossmarks.fridalab -l challenge8.js
```

```
jind>frida -U -f com.ad2001.frida0x8
```

```
Frida 16.1.5 - A world-class dynamic instrumentation toolkit
```

```
Commands:
```

```
  help      -> Displays the help system  
  object?   -> Display information about 'object'  
  exit/quit -> Exit
```

```
More info at https://frida.re/docs/home/
```

```
Connected to Android Emulator 5554 (id=emulator-5554)
```

```
com.ad2001.frida0x8`. Resuming main thread!
```

```
emulator 5554::com.ad2001.frida0x8 ]->
```

Final output look likes :

